

Physics Note

Matsumotofukashi High School

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form 2026-04-14

1 *Electric Field*

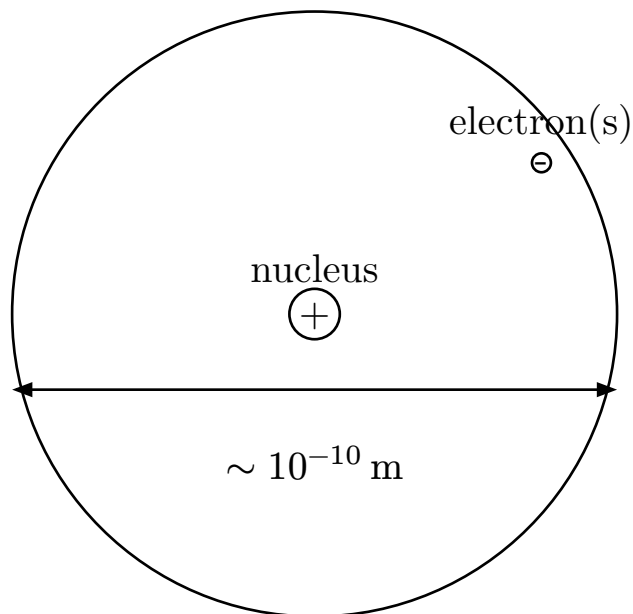
1.1 Electrostatic Force

- **electrification** - A process getting charge
- **static electoricity** - static charge
- **electoric charge** -
- **point charge** - charge which can be ignored the size

Electrostatic force is an interraction between charged particles.

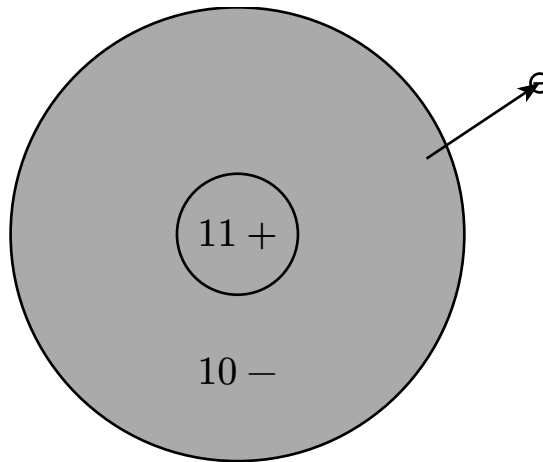
$$|\mathbb{F}| = k \frac{q_1 q_2}{r^2}$$

1.2 Structure of Atom



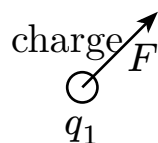
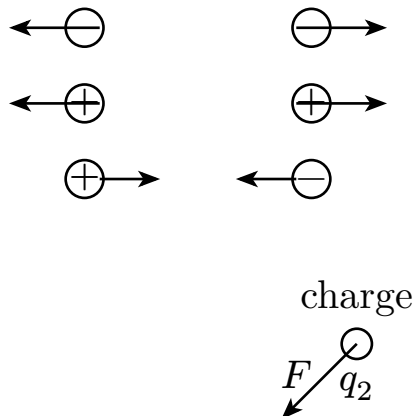
- **elementary charge** $1.6 \cdot 10^{19} \text{ C}$

1.3 Mechanism of Electrification



Total amount of charge is conserved.

– **Coulomb's law**



$$F = k \frac{q_1 q_2}{r^2}$$

k depends on type of surrounding substance.

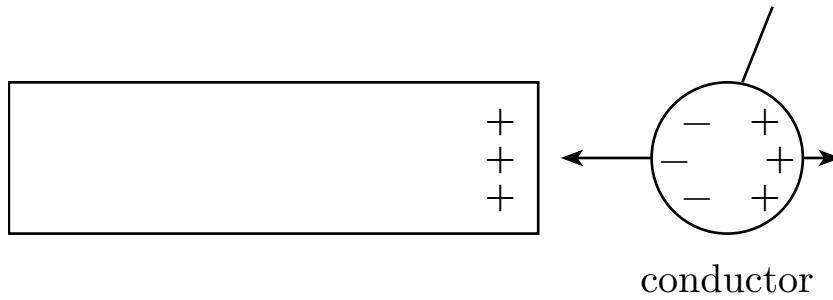
In vacuum, $k \approx 9.0 \cdot 10^9 \text{ N m}^2 \text{ C}^{-2}$

1.4 Electrostatic Induction

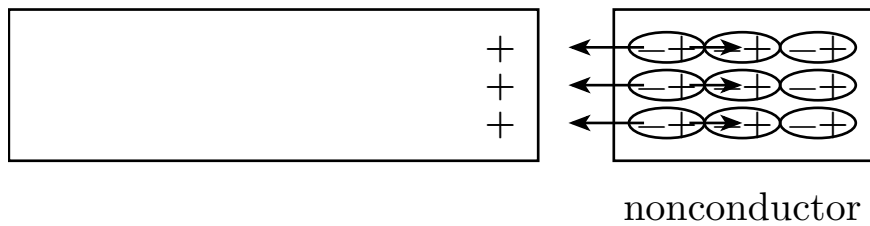
– **Conductor** - Substances which can conduct electricity.

ex) metal, carbon

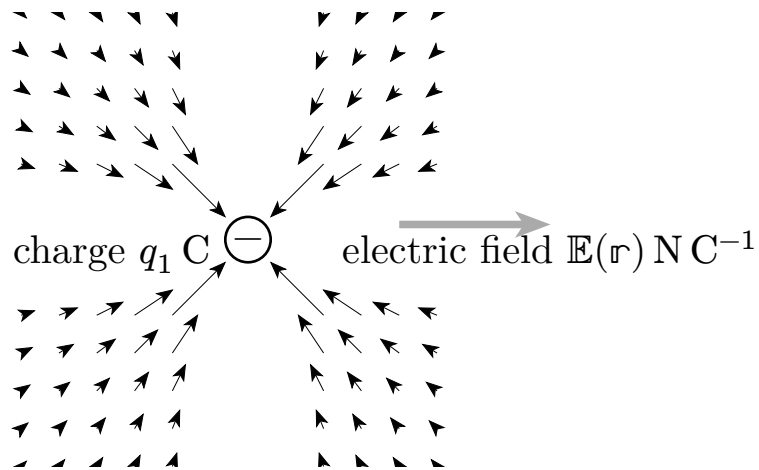
- **Nonconductor** - Substances which cannot conduct electricity.
- **Semiconductor** - Substances which have middle electric resistance between conductor and nonconductor.
- **induction**



- **Dielectric Polarization**



1.5 Electric Field



Electric field is vector field.

Let 1 C charge be a test charge at position $\mathbf{r}$, $\mathbb{E}(\mathbf{r})$ is the electric force which the test charge receive.

$$\mathbb{E}(\mathbf{r}) = kQ \frac{\mathbf{r}}{|\mathbf{r}|^3}$$

If there are multiple charges, the electric field is equals to sum of each electric fields made by these charges.

1.6 Electric line of force

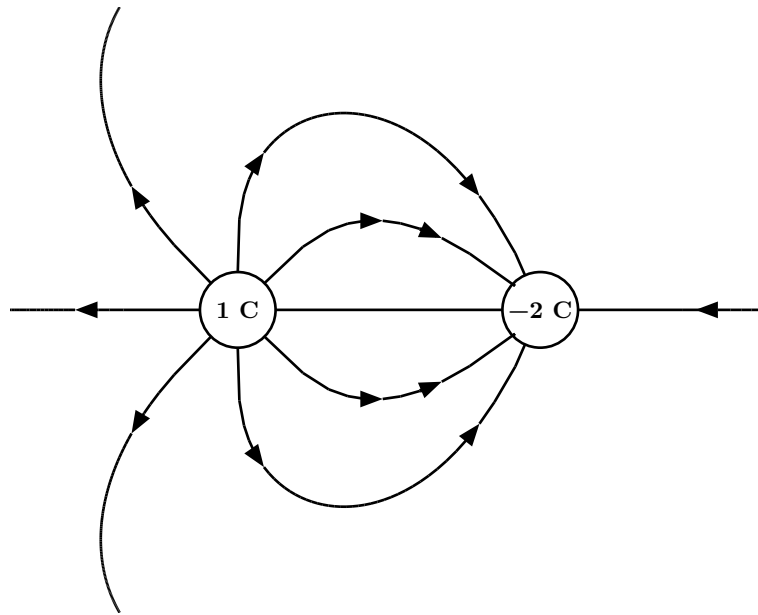
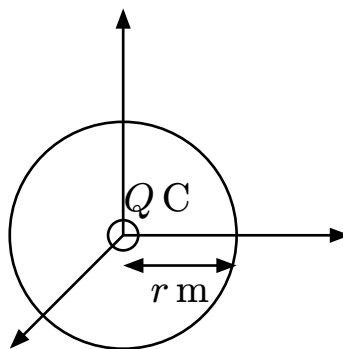


Figure 1: simulator

(it was difficult to draw the figure...)

- the direction of tangent of line is the same as the direction of the field's vector.
- the line appears with plus charge and disappear with minus charge.
- E lines drawn per 1 m^2 , as strength of electric field is $E \text{ N C}^{-1}$.



Let N be number of lines, Q amount of charge.

$$N = E \cdot 4\pi r^2 = k \frac{Q}{r^2} \cdot 4\pi r^2$$

Thus,

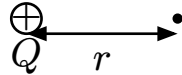
$$N = 4\pi kQ$$

1.7 Electric Potential

Let U potential energy of charge which have q , V electric potential,

$$V := \frac{U}{q}$$

Thus, V is potential energy per unit charge.



Let infinity be reference level of potential,

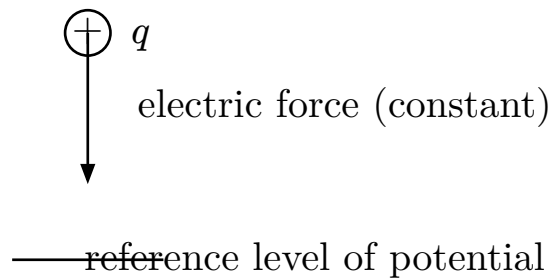
$$V(r) = \int_r^\infty dr \cdot k \frac{Q}{r^2} = kQ[-r^{-1}]_r^\infty = k \frac{Q}{r}$$

$$V(r) = k \frac{Q}{r}$$



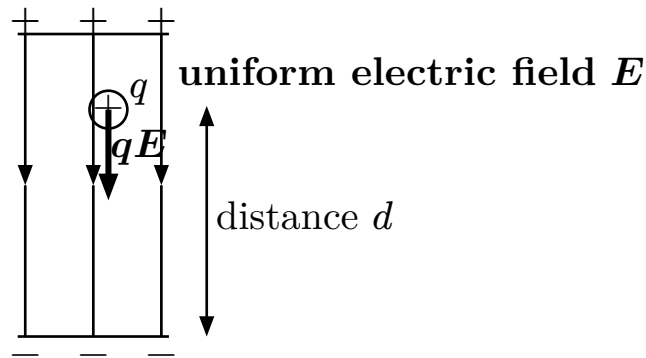
$$V_{AB} = V_A - V_B \text{ (reference is } B)$$

V_{AB} is also called **voltage**.



$$W_{AB} = q(V_A - V_B)$$

1.8 Relative of Field and Potential



$$qEd = qV$$

$$\Leftrightarrow V = Ed$$

electric potential is proportional to energy, so it's superposition is calculated by sum.

1.9 Equipotential Surface

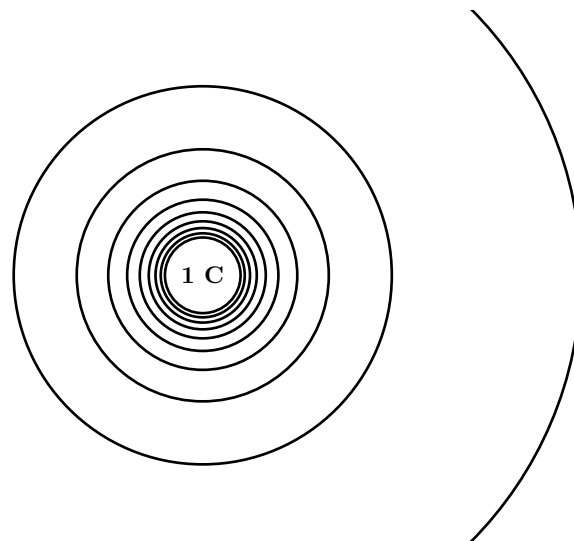


Figure 2: simulator

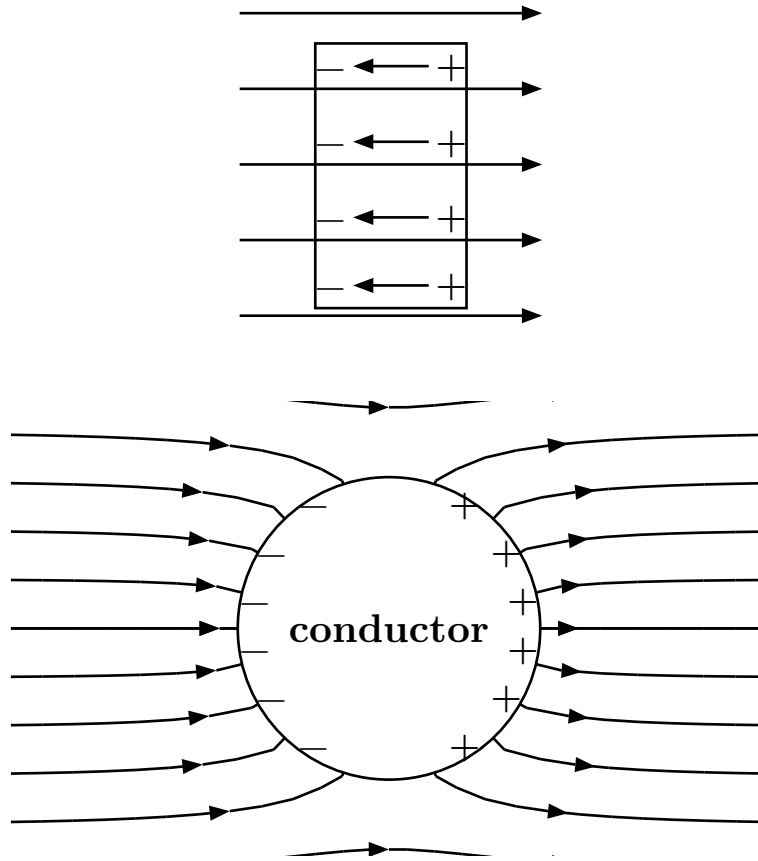
- **Equipotential Surface** - A surface which was made by joining points that have equal electric potential.

equipotential surface is perpendicular to electric line of force.

(if test charge move with a direction perpendicular to electric force , work is $W = \mathbb{F} \cdot \Delta \mathbb{x} \Leftrightarrow$ potential is same constantly.)

$$\frac{1}{2}mv^2 + qV = \text{const.}$$

1.10 Substance and Electric Field



In conductor, electric field is 0. (charges continue moving while field is not 0,)

It means that electric potential is the same in metal.

- **Electric field is perpendicular to surface of metal.** (metal distorts surrounding electric field)
- **Electric charges do not appear in internal of metal.** (on surface only)

(why? is not impossible to make field 0 with it having charges in metal? そうかな...そうかも...引力と斥力があるから自動的に表面にでできるかね。べつに全電荷が打ち消しに貢献する必要はなく、正負が surface に交互に并んでいればいい?)